Graduate Level Elementary Particle Physics

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Contents

*https://github.com/yuriphestos/gradphestos-epp

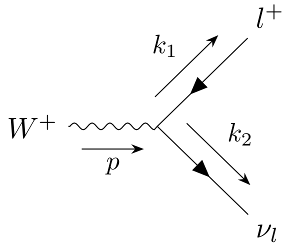
1 Task 1

1.1 Decay Width of W and Z Bosons

Calculate the width for the decay of the W and Z bosons, neglecting $m_f/m_{W(Z)}$. Compare with the PDG. What is the branching ratio of Z boson decay to neutrinos? How can you measure this branching ratio at e^+e^- collider that produces Z bosons resonantly, provided that the final state neutrinos are not observed?

Decay width of W boson:

$$W^+ \longrightarrow l^+\nu_l$$



The matrix element is given by

$$\mathcal{M} = \frac{g}{2\sqrt{2}}\,\varepsilon_\mu \bar{u}_\nu \gamma^\mu (1-\gamma^5) v_l$$

Summing over the polarization states and fermion spins, we obtain

$$\begin{aligned}
\left\langle |\mathcal{M}|^2 \right\rangle &= \frac{1}{3}\cdot\frac{g^2}{8}\left(\frac{p_\mu p_\nu}{m_W^2} - g_{\mu\nu}\right)\mathrm{Tr}\left[\not{k}_2\gamma^\mu(1-\gamma^5)\not{k}_1\gamma^\nu(1-\gamma^5)\right] \\
&= \frac{g^2}{3}\left[\frac{2(k_1\cdot p)(k_2\cdot p)}{m_W^2} + k_1\cdot k_2\right]
\end{aligned}$$

Neglecting m_f/m_W, this reduces to $g^2 m_W^2/3$. The decay width can be calculated as follows:

$$\begin{aligned}
\int \mathrm{d}\Gamma_W &= \int \frac{1}{2m_W}\frac{\mathrm{d}^3k_1}{(2\pi)^3}\frac{1}{2E_1}\frac{\mathrm{d}^3k_2}{(2\pi)^3}\frac{1}{2E_2}\left|\mathcal{M}\right|^2(2\pi)^4\delta^4(p-k_1-k_2) \\
&= \int \frac{1}{2m_W}\frac{\mathrm{d}^3k_1}{(2\pi)^3}\frac{1}{2E_1}\frac{\mathrm{d}^3k_2}{(2\pi)^3}\frac{1}{2E_2}\left|\mathcal{M}\right|^2(2\pi)^4\delta(E_W-E_1-E_2)\delta^3(\vec{p}-\vec{k}_1-\vec{k}_2) \\
&= \frac{g^2 m_W}{96\pi^2}\int\frac{\mathrm{d}^3k_1\mathrm{d}^3k_2}{|\vec{k}_1||\vec{k}_2|}\delta(m_W-|\vec{k}_1|-|\vec{k}_2|)\delta^3(\vec{p}-\vec{k}_1-\vec{k}_2) \\
&= \frac{g^2 m_W}{96\pi^2}\int\frac{\mathrm{d}^3k_2}{|\vec{p}-\vec{k}_2||\vec{k}_2|}\delta(m_W-|\vec{p}-\vec{k}_2|-|\vec{k}_2|) \\
&= \frac{g^2 m_W}{96\pi^2}\int\frac{k_2\mathrm{d}k_2\mathrm{d}\Omega}{m_W-k_2}\delta(m_W-|\vec{p}-\vec{k}_2|-k_2)
\end{aligned}$$

where $|\vec{p}-\vec{k}_2| = \sqrt{p^2+k_2^2-2pk_2\cos\theta}$, $m_W \gg \vec{p}$ and $m_f \to 0$. Using the properties

$$\delta(f(k_2)) = \frac{\delta(k_2-k_0)}{|f'(k_0)|}\,; \qquad f(k_0)=0$$

$$\Gamma_W = \frac{g^2 m_W}{192\pi^2}\int \frac{k_2 \mathrm{d}k_2 \mathrm{d}\Omega}{m_W - k_2}\delta\left(k_2 - \frac{m_W}{2}\right) = \frac{g^2 m_W}{48\pi} = \frac{G_F m_W^3}{6\sqrt{2}\pi}$$

Thus,

$$\Gamma(W^+ \to l^+\nu_l) = \frac{G_F m_W^3}{6\sqrt{2}\pi} \simeq 227 \text{ MeV}$$

where

$$m_W = 80.377 \text{ GeV} \; ; \qquad G_F = 1.1663788 \times 10^{-5} \text{ GeV}^{-2}$$

Similarly for quarks, the CKM matrix element V_{ij} replaces the leptonic vertex. We use the two-generation Cabibbo approximation ($V_{ud} \simeq V_{cs} \simeq \cos\theta_c$, $V_{us} \simeq -V_{cd} \simeq \sin\theta_c$), which is valid since the third-generation entries $|V_{ub}|$, $|V_{cb}|$ are very small. Note that the decay $W^+ \to t\bar{b}$ is kinematically forbidden since $m_t > m_W$.

$$W^+(p) \longrightarrow \; u(k_1)\bar{d}(k_2), \; c(k_1)\bar{s}(k_2)$$

The matrix element for both is

$$\mathcal{M} = \frac{gV_{ij}}{2\sqrt{2}}\,\varepsilon_\mu \bar{u}_u \gamma^\mu(1-\gamma^5)v_d \quad \Longrightarrow \quad |\mathcal{M}|^2 = g^2 m_W^2 |V_{ij}|^2$$

The calculation is the same but we have color factor $N_c = 3$ for quarks and CKM elements.

$$\Gamma(W^+ \to u\bar{d}\,) = \Gamma(W^+ \to c\bar{s}) = \frac{G_F m_W^3 \cos^2\theta_c}{2\sqrt{2}\pi} \simeq 647 \text{ MeV}$$

The Cabibbo-suppressed modes give

$$\Gamma(W^+ \to u\bar{s}) = \Gamma(W^+ \to c\bar{d}\,) = \frac{G_F m_W^3 \sin^2\theta_c}{2\sqrt{2}\pi} \simeq 34.6 \text{ MeV}$$

Note that by CKM unitarity, $\sum_j |V_{ij}|^2 = 1$ for each row, so summing over all kinematically accessible quark channels:

$$\sum_{i=u,c}\sum_{j=d,s,b}|V_{ij}|^2 = 2$$

The total hadronic width is thus $\Gamma(W \to \text{had}) = 2N_c\, G_F m_W^3/(6\sqrt{2}\pi)$, independent of the specific CKM values. Now comparing with PDG:

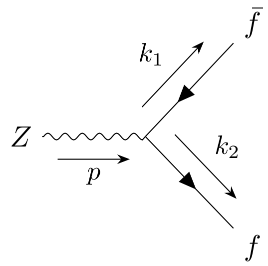
$$\Gamma_W = 3\Gamma(W^+ \to l^+\nu_l) + 2\Gamma(W^+ \to u\bar{d}\,) + 2\Gamma(W^+ \to u\bar{s}) \simeq 2.044 \text{ GeV}$$

$$\frac{\Gamma_W}{\Gamma_{W,PDG}} = \frac{2.044}{2.085} = 0.980$$

The ∼2 % discrepancy is expected, as our tree-level result neglects $\mathcal{O}(\alpha_s/\pi) \sim 3$–$4$ % QCD corrections to the hadronic decay channels. Including the leading QCD correction factor $(1 + \alpha_s(m_W)/\pi)$ to the quark modes would improve the agreement.

Decay width of Z boson:

$$Z \longrightarrow \; \bar{f}f$$



$$\mathcal{M} = \frac{g}{2\cos\theta_w}\,\varepsilon_\mu \bar{u}_f \gamma^\mu (g_V - g_A\gamma^5) v_f$$

Summing over the polarization states and fermion spins, we obtain

$$\begin{aligned}
\left\langle |\mathcal{M}|^2 \right\rangle &= \frac{1}{3}\cdot\frac{g^2}{4\cos^2\theta_w}\left(\frac{p_\mu p_\nu}{m_Z^2} - g_{\mu\nu}\right)\mathrm{Tr}\left[\not{k}_2\gamma^\mu(g_V - g_A\gamma^5)\not{k}_1\gamma^\nu(g_V - g_A\gamma^5)\right] \\
&= \frac{g^2}{3\cos^2\theta_w}\left(g_V^2 + g_A^2\right)\left[\frac{2(k_1\cdot p)(k_2\cdot p)}{m_Z^2} + k_1\cdot k_2\right] \\
&\simeq \frac{g^2 m_Z^2}{3\cos^2\theta_w}\left(g_V^2 + g_A^2\right)
\end{aligned}$$

Since we are taking the limit $m_f/m_{Z(W)} \to 0$, the decay width formula takes the same form as for the W boson:

$$\Gamma_Z = \frac{g^2 m_Z^2}{48\pi\cos^2\theta_w}\left(g_V^2 + g_A^2\right) = \frac{G_F m_Z^3}{6\sqrt{2}\pi}\left(g_V^2 + g_A^2\right)$$

where $m_Z = 91.1876$ GeV and $\sin^2\theta_w = 0.2312$.

$$\begin{aligned}
\Gamma(Z \to \bar{\nu}_l \nu_l) &= \frac{G_F m_Z^3}{12\sqrt{2}\pi} \simeq 166 \text{ MeV} \\
\Gamma(Z \to l^+ l^-) &= \frac{G_F m_Z^3}{6\sqrt{2}\pi}\left[\left(-\frac{1}{2} + 2\sin^2\theta_w\right)^2 + \left(\frac{1}{2}\right)^2\right] \simeq 83.4 \text{ MeV}
\end{aligned}$$

For the quark channels, an additional color factor $N_c = 3$ is required:

$$\begin{aligned}
\Gamma(Z \to \bar{u}u) = \Gamma(Z \to \bar{c}c) &= \frac{G_F m_Z^3}{2\sqrt{2}\pi}\left[\left(\frac{1}{2} - \frac{4}{3}\sin^2\theta_w\right)^2 + \left(\frac{1}{2}\right)^2\right] \simeq 286 \text{ MeV} \\
\Gamma(Z \to \bar{d}d) = \Gamma(Z \to \bar{s}s) = \Gamma(Z \to \bar{b}b) &= \frac{G_F m_Z^3}{2\sqrt{2}\pi}\left[\left(-\frac{1}{2} + \frac{2}{3}\sin^2\theta_w\right)^2 + \left(\frac{1}{2}\right)^2\right] \simeq 368 \text{ MeV}
\end{aligned}$$

Thus,

$$\Gamma_Z = 3\Gamma(Z \to \bar{\nu}_l\nu_l) + 3\Gamma(Z \to l^+l^-) + 2\Gamma(Z \to \bar{u}u) + 3\Gamma(Z \to \bar{d}d) \simeq 2.4242 \text{ GeV}$$

$$\frac{\Gamma_Z}{\Gamma_{Z,PDG}} = \frac{2.4242}{2.4952} = 0.972$$

As with the W width, the ∼3 % deficit is due to the absence of $\mathcal{O}(\alpha_s/\pi)$ QCD corrections in the hadronic channels. The branching ratio of Z boson decay to neutrinos (per flavor) is

$$\frac{\Gamma(Z \to \bar{\nu}_l\nu_l)}{\Gamma_Z} \times 100\,\% = 6.85\,\%$$

The total invisible branching ratio, summing over all three neutrino species, is

$$\frac{\Gamma_{\text{inv}}}{\Gamma_Z} = \frac{3\Gamma(Z \to \bar{\nu}_l \nu_l)}{\Gamma_Z} = \frac{3 \times 166}{2424} = 20.5\,\%$$

This is in good agreement with the PDG value of $\Gamma_{\text{inv}}/\Gamma_Z = 20.00 \pm 0.06\%$.

Measuring the neutrino branching ratio at e^+e^- colliders:

At an e^+e^- collider operating at the Z resonance (as at LEP), neutrinos in the final state escape the detector unobserved. However, the invisible width can be extracted indirectly using the Z resonance lineshape. By scanning the center-of-mass energy $\sqrt{s}$ around m_Z, one measures the total hadronic and leptonic cross sections, which determine the peak cross section σ^0_{peak}, the total width Γ_Z, and the visible partial widths Γ_{had}, Γ_{ll}. The invisible width is then obtained by subtraction:

$$\Gamma_{\text{inv}} = \sum_l \Gamma(Z \to \bar{\nu}_l \nu_l) = \Gamma_Z - \Gamma_{\text{had}} - 3\Gamma_{ll}$$

Equivalently, using the peak hadronic cross section at the Z pole:

$$\sigma^0_{\text{had}} = \frac{12\pi}{m_Z^2} \frac{\Gamma_{ee}\Gamma_{\text{had}}}{\Gamma_Z^2}$$

one can extract Γ_Z from σ^0_{had} together with the measured partial widths. The ratio $\Gamma_{\text{inv}}/\Gamma_{ll}$ then gives the number of light neutrino species:

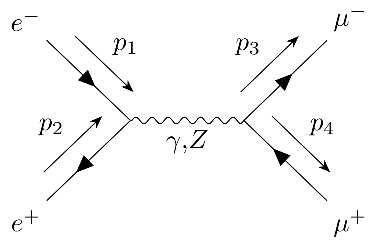
$$N_\nu = \frac{\Gamma_{\text{inv}}}{\Gamma(Z \to \bar{\nu}_l \nu_l)_{\text{SM}}} = 2.9840 \pm 0.0082 \quad \text{(PDG)}$$

This is entirely consistent with $N_\nu = 3$.

1.2 Total Cross Section of $e^+e^- \to \mu^+\mu^-$

Calculate the total cross section for the tree-level $e^+e^- \to \mu^+\mu^-$ process mediated by γ, Z bosons. Neglect m_e, μ, assume that initial states are not polarized. Observe that a naive answer diverges when $s \to m_Z$.

$$e^+e^- \longrightarrow \mu^+\mu^-$$



The matrix elements are:

$$\mathcal{M}_\gamma = \bar{v}_2(e\gamma^\mu)u_1\left(\frac{g_{\mu\nu}}{q^2}\right)\bar{u}_3(e\gamma^\nu)v_4 = \frac{e^2}{q^2}(\bar{v}_2\gamma^\mu u_1)(\bar{u}_3\gamma_\mu v_4)$$

$$\mathcal{M}_Z = \frac{g^2}{4\cos^2\theta_w}\,\bar{v}_2\gamma^\mu(g_V - g_A\gamma^5)u_1\left(\frac{g_{\mu\nu} - \frac{q_\mu q_\nu}{m_Z^2}}{q^2 - m_Z^2}\right)\bar{u}_3\gamma^\nu(g_V - g_A\gamma^5)v_4$$

$$|\mathcal{M}_\gamma + \mathcal{M}_Z|^2 = |\mathcal{M}_\gamma|^2 + |\mathcal{M}_Z|^2 + 2\mathcal{M}_\gamma^*\mathcal{M}_Z$$

Summing over all final-state spins and averaging over initial-state spins ($m_{e,\mu} \to 0$):

$$\frac{1}{4}\sum_{\text{spin}}|\mathcal{M}_\gamma|^2 = \frac{e^4}{4q^4}\mathrm{Tr}\left[\not{p}_1\gamma^\nu\not{p}_2\gamma^\mu\right]\mathrm{Tr}\left[\not{p}_3\gamma_\mu\not{p}_4\gamma_\nu\right]$$

Using standard trace identities and the Mandelstam variables:

$$\begin{aligned}\mathrm{Tr}\left[\not{p}_1\gamma^\nu\not{p}_2\gamma^\mu\right] &= p_1^\alpha p_2^\beta\,\mathrm{Tr}\left[\gamma^\alpha\gamma^\nu\gamma^\beta\gamma^\mu\right] = 4p_1^\alpha p_2^\beta\left(g^{\alpha\nu}g^{\beta\mu} - g^{\alpha\beta}g^{\nu\mu} + g^{\alpha\mu}g^{\nu\beta}\right)\\ &= 4\left(p_1^\nu p_2^\mu + p_1^\mu p_2^\nu - p_1^\alpha p_2^\alpha g^{\nu\mu}\right)\end{aligned}$$

$$\begin{aligned}s &= (p_1+p_2)^2 = (p_3+p_4)^2 = 2p_1\cdot p_2 = 2p_3\cdot p_4 = q^2\\ t &= (p_1-p_3)^2 = (p_2-p_4)^2 = -2p_1\cdot p_3 = -2p_2\cdot p_4\\ u &= (p_1-p_4)^2 = (p_2-p_3)^2 = -2p_1\cdot p_4 = -2p_2\cdot p_3\end{aligned}$$

where

$$p_1 = (E,\vec{p}_1)\,;\qquad p_2 = (E,-\vec{p}_1)\,;\qquad p_3 = (E,\vec{p}_3)\,;\qquad p_4 = (E,-\vec{p}_3)$$
$$s = 4E^2 = E_{cm}^2\,;\quad t = -2E^2 + 2|\vec{p}_1||\vec{p}_3|\cos\theta\,;\quad u = -2E^2 - 2|\vec{p}_1||\vec{p}_3|\cos\theta$$

$$\begin{aligned}\left\langle|\mathcal{M}_\gamma|^2\right\rangle &= \frac{8e^4}{q^4}\left[(p_1\cdot p_4)(p_2\cdot p_3) + (p_1\cdot p_3)(p_2\cdot p_4)\right] = \frac{2e^4}{s^2}(u^2+t^2)\\ &= e^4(1+\cos^2\theta)\end{aligned}$$

Similarly, $\not{q} = \not{p}_1 + \not{p}_2 = \not{p}_3 + \not{p}_4$ and $\bar{u}\not{p} = 0$ so $\bar{u}_i(\not{p}_i - \not{p}_j)u_j = 0$. Defining $g_Z \equiv g/\cos\theta_w$, the $q_\mu q_\nu/m_Z^2$ term in the Z propagator vanishes by the equations of motion, giving

$$\mathcal{M}_Z = \frac{g_Z^2}{4\,(q^2 - m_Z^2)} \left[\bar{v}_2\gamma^\mu(g_V - g_A\gamma^5)u_1\right] \left[\bar{u}_3\gamma_\mu(g_V - g_A\gamma^5)v_4\right]$$

$$\begin{aligned}
\left\langle|\mathcal{M}_Z|^2\right\rangle &= \frac{1}{4}\cdot\frac{g_Z^4}{16\,(q^2-m_Z^2)^2}\cdot 32\left[\left(g_V^2-g_A^2\right)^2(p_1\cdot p_3)\,(p_2\cdot p_4) + \left(g_V^4+6g_V^2g_A^2+g_A^4\right)(p_1\cdot p_4)\,(p_2\cdot p_3)\right]\\
&= \frac{g_Z^4 s^2}{32\,(s-m_Z^2)^2}\left[\left(g_V^2-g_A^2\right)^2(1-\cos\theta)^2 + \left(g_V^4+6g_V^2g_A^2+g_A^4\right)(1+\cos\theta)^2\right]
\end{aligned}$$

$$\begin{aligned}
\left\langle\mathcal{M}_\gamma^*\mathcal{M}_Z\right\rangle &= \frac{e^2g_Z^2}{4q^2(q^2-m_Z^2)}\cdot 8\left[(g_V^2-g_A^2)\,(p_1\cdot p_3)\,(p_2\cdot p_4) + (g_V^2+g_A^2)\,(p_1\cdot p_4)\,(p_2\cdot p_3)\right]\\
&= \frac{e^2g_Z^2 s}{8(s-m_Z^2)}\left[(g_V^2-g_A^2)(1-\cos\theta)^2 + (g_V^2+g_A^2)(1+\cos\theta)^2\right]
\end{aligned}$$

Integrating the differential cross section over the solid angle:

$$\begin{aligned}
\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2}\frac{1}{s} = \frac{1}{64\pi^2 s}\Bigg\{&e^4(1+\cos^2\theta) + \frac{g_Z^4s^2}{32\,(s-m_Z^2)^2}\Big[\left(g_V^2-g_A^2\right)^2(1-\cos\theta)^2\\
&+\left(g_V^4+6g_V^2g_A^2+g_A^4\right)(1+\cos\theta)^2\Big] + \frac{e^2g_Z^2s}{4(s-m_Z^2)}\left[(g_V^2-g_A^2)(1-\cos\theta)^2+(g_V^2+g_A^2)(1+\cos\theta)^2\right]\Bigg\}
\end{aligned}$$

$$\boxed{\sigma(e^+e^-\to\mu^+\mu^-) = \left[\frac{4\pi\alpha^2}{3s}\right]_\gamma + \left[\frac{g_Z^4 s\left(g_V^2+g_A^2\right)^2}{192\pi\,(s-m_Z^2)^2}\right]_Z + \left[\frac{\alpha g_Z^2 g_V^2}{12(s-m_Z^2)}\right]_I}$$

where $\alpha = e^2/4\pi$, $g_Z = g/\cos\theta_w$, $g_V = 2\sin^2\theta_w - 1/2$ and $g_A = -1/2$. Note that the interference term contains only g_V^2 (not $g_V^2 + g_A^2$) because the axial-vector contribution is odd in $\cos\theta$ and vanishes upon angular integration for the total cross section; it does, however, produce a forward-backward asymmetry A_{FB}. One can see that the result diverges at $s \to m_Z^2$.

1.3 Near-Resonance Cross Section

What is the correct generalization of the formula that makes σ finite near the resonance? Consider $m_Z \to m_Z - \frac{i}{2}\Gamma_Z$ substitution. Show that the near-resonance cross section agrees with the Breit-Wigner formula (Eq. (51.1), "Cross-Section Formulae for Specific Processes" review on PDG). Keeping the number of neutrinos as free parameter (2, 3, 4...) show that you can reproduce Figure 53.2 from the PDG, that was used by LEP scientists to determine that the number of "active" neutrino species is three.

Consider the previous task with Z exchange only (dominant at Z resonance). We now modify the divergence at the pole $\sqrt{s} = m_Z$ by substituting $m_Z \to m_Z - \frac{i}{2}\Gamma_Z$, where Γ_Z is the total width from Section 1.1. In the propagator denominator:

$$s - m_Z^2 \;\longrightarrow\; s - \left(m_Z - \frac{i}{2}\Gamma_Z\right)^2 = s - m_Z^2 + i m_Z \Gamma_Z + \frac{\Gamma_Z^2}{4}$$

Defining $\bar{s} \equiv s - m_Z^2 + \Gamma_Z^2/4$, the squared propagator becomes

$$\frac{1}{|s - m_Z^2 + i m_Z \Gamma_Z + \Gamma_Z^2/4|^2} = \frac{1}{\bar{s}^2 + m_Z^2 \Gamma_Z^2}$$

The Z-exchange cross section from the previous section (dropping the $\Gamma_Z^2/4$ mass shift which is higher-order in Γ_Z/m_Z) is

$$\sigma(e^+e^- \to f\bar{f})\Big|_Z \simeq \frac{g_Z^4 s \left(g_V^2 + g_A^2\right)^2}{192\pi\left[(s - m_Z^2)^2 + m_Z^2\Gamma_Z^2\right]}$$

At the resonance $s = m_Z^2$, substituting $g_Z = g/\cos\theta_w$ and using $G_F = g^2/(4\sqrt{2}m_W^2)$, $m_W = m_Z\cos\theta_w$:

$$\sigma_{\text{peak}} = \frac{g_Z^4 m_Z^2}{192\pi\Gamma_Z^2}\left(g_V^2 + g_A^2\right)^2$$

Recalling from Section 1.1 that $\Gamma(Z \to f\bar{f}) = G_F m_Z^3/(6\sqrt{2}\pi)(g_V^{f\,2} + g_A^{f\,2})$, one can verify the identity:

$$\frac{g_Z^4 (g_V^{e\,2} + g_A^{e\,2})(g_V^{f\,2} + g_A^{f\,2})}{192\pi} = \frac{12\pi\,\Gamma_{ee}\Gamma_{ff}}{m_Z^2}$$

and therefore the cross section can be also written as:

$$\boxed{\sigma(e^+e^- \to f\bar{f}) = \frac{12\pi s}{m_Z^2} \cdot \frac{\Gamma_{ee}\,\Gamma_{ff}}{(s - m_Z^2)^2 + m_Z^2\Gamma_Z^2}}$$

$$\sigma(e^+e^- \to f\bar{f})\Big|_{s \to m_Z^2} = \frac{12\pi}{m_Z^2} \cdot \frac{\Gamma_{ee}\,\Gamma_{ff}}{\Gamma_Z^2}$$

This is the relativistic Breit-Wigner formula (PDG Eq. 51.1). A more precise form, commonly adopted in LEP analyses, uses an s-dependent total width in the denominator to better capture the off-peak behavior of a spin-1 resonance:

$$\sigma(e^+e^- \to f\bar{f}) = \frac{12\pi\, s}{m_Z^2} \cdot \frac{\Gamma_{ee}\,\Gamma_{ff}}{(s - m_Z^2)^2 + s^2\,\Gamma_Z^2/m_Z^2}$$

At the peak ($s = m_Z^2$) both forms coincide, giving the pole cross section $\sigma_f^0 = 12\pi\,\Gamma_{ee}\Gamma_{ff}/(m_Z^2\,\Gamma_Z^2)$.

Number of neutrino species and initial-state radiation:

The iconic measurement of the number of light, active neutrino species at LEP relies on the sensitive dependence of the Z resonance lineshape on the total width. As computed in Section 1.1:

$$\Gamma_Z = \Gamma_{\rm had} + 3\,\Gamma_{ll} + N_\nu\,\Gamma_{\nu\bar{\nu}}$$

where N_ν is the number of neutrino flavors. Since $\Gamma_{\nu\bar{\nu}} \simeq 166$ MeV, each additional species increases Γ_Z by 166 MeV. Because the peak hadronic cross section scales as

$$\sigma^0_{\rm had} = \frac{12\pi}{m_Z^2}\,\frac{\Gamma_{ee}\,\Gamma_{\rm had}}{\Gamma_Z^2}$$

additional neutrino species suppress the peak height ($\sigma^0 \propto 1/\Gamma_Z^2$) and broaden the resonance.

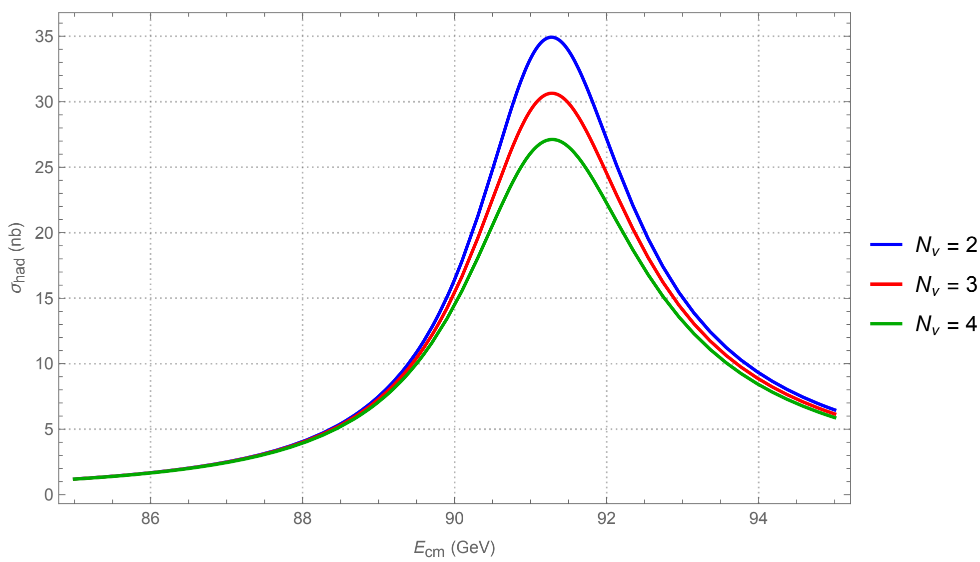
However, to reproduce the *observed* lineshape in PDG Figure 53.2, it is not sufficient to plot the bare Breit-Wigner: the measured cross section is distorted by **initial-state radiation** (ISR). The incoming $e^\pm$ radiate collinear photons before the hard scattering, reducing the effective center-of-mass energy. The observed cross section is a convolution of the Born-level cross section with the ISR radiator function:

$$\sigma_{\rm obs}(s) = \int_0^1 {\rm d}x\; G(x,s)\;\sigma_{\rm BW}(s(1-x))$$

where x is the fractional energy radiated and $G(x,s)$ is the QED structure function. At leading-logarithmic order with soft/virtual corrections (Kuraev–Fadin, 1985):

$$G(x,s) = \beta x^{\beta-1}\left(1+\frac{3\beta}{4}\right) - \beta\left(1-\frac{x}{2}\right) + \mathcal{O}(\beta^2)\,, \qquad \beta = \frac{2\alpha}{\pi}\left(\ln\frac{s}{m_e^2} - 1\right)$$

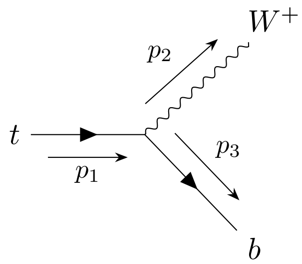
At the Z pole, $\beta \simeq 0.108$. The singular $x^{\beta-1}$ factor concentrates the radiation at small x (soft photons), while the remaining terms account for hard collinear emission. The figure below, generated numerically using the Kuraev–Fadin convolution with PDG partial widths (see `task-1-3.m`), reproduces Figure 53.2 of the PDG:



1.4 Decay Rate of $t \to W^+b$

Calculate the decay rate of the top quark to W^+b pairs. Consider the limit of $m_t/m_W \gg 1$. Can you interpret the enhancement of the Γ_t in this limit using the equivalence theorem? Entertain a possibility of an imaginary world described by the SM with exactly the same Higgs expectation value and the rest of the parameters, but with $g_W \to 0$. Is electron stable in this limit?

$$t \longrightarrow W^+b$$



The matrix element is:

$$\mathcal{M} = \frac{g}{2\sqrt{2}}\,\bar{u}_3\gamma^\mu(1-\gamma^5)u_1\varepsilon^*_\mu$$

$$\begin{aligned}
\langle|\mathcal{M}|^2\rangle &= \frac{1}{2}\cdot\frac{g^2}{8}\cdot 8\,(p_3^\mu p_1^\nu + p_3^\nu p_1^\mu - g^{\mu\nu}p_3\cdot p_1)\left(-g_{\mu\nu} + \frac{p_{2,\mu}p_{2,\nu}}{m_W^2}\right)\\
&= \frac{g^2}{2}\left[\frac{2(p_1\cdot p_2)(p_2\cdot p_3)}{m_W^2} + p_1\cdot p_3\right]\\
&= \frac{g^2}{2}\left[\frac{(m_t^2+m_W^2-m_b^2)(m_t^2-m_W^2-m_b^2)}{2m_W^2} + \frac{m_t^2-m_W^2+m_b^2}{2}\right]\\
&= \frac{g^2m_W^2}{2}\left(\frac{m_t^2}{m_W^2}-1\right)\left(\frac{m_t^2}{2m_W^2}+1\right)\\
&\simeq \frac{g^2m_t^4}{4m_W^2}
\end{aligned}$$

where $m_b \to 0$, $m_t/m_W \gg 1$

$$p_1\cdot p_2 = \frac{m_t^2+m_W^2-m_b^2}{2}\,;\quad p_2\cdot p_3 = \frac{m_t^2-m_W^2-m_b^2}{2}\,;\quad p_1\cdot p_3 = \frac{m_t^2-m_W^2+m_b^2}{2}$$

Decay Rate:

$$\begin{aligned}
\int \mathrm{d}\Gamma_t &= \int \frac{1}{2m_t}\frac{\mathrm{d}^3p_W}{(2\pi)^3}\frac{1}{2E_W}\frac{\mathrm{d}^3p_b}{(2\pi)^3}\frac{1}{2E_b}\,|\mathcal{M}|^2\,(2\pi)^4\delta(E_t-E_W-E_b)\delta^3(\vec{p}_t-\vec{p}_W-\vec{p}_b)\\
&= \frac{g^2m_t^3}{128\pi^2m_W^2}\int\frac{\mathrm{d}^3p_W}{\sqrt{p_W^2+m_W^2}}\frac{\mathrm{d}^3p_b}{|\vec{p}_b|}\,\delta(\sqrt{p_t^2+m_t^2}-\sqrt{p_W^2+m_W^2}-|\vec{p}_b|)\delta^3(\vec{p}_t-\vec{p}_W-\vec{p}_b)\\
&= \frac{g^2m_t^3}{128\pi^2m_W^2}\int\frac{p_b^2\mathrm{d}p_b\mathrm{d}\Omega}{|\vec{p}_b|\sqrt{|\vec{p}_t-\vec{p}_b|^2+m_W^2}}\,\delta(\sqrt{p_t^2+m_t^2}-\sqrt{|\vec{p}_t-\vec{p}_b|^2+m_W^2}-|\vec{p}_b|)\\
&= \frac{g^2m_t^3}{128\pi^2m_W^2}\int\frac{p_b^2\mathrm{d}p_b\mathrm{d}\Omega}{|\vec{p}_b|(m_t-|\vec{p}_b|)}\,\delta(m_t-\sqrt{|\vec{p}_b|^2+m_W^2}-|\vec{p}_b|)\\
&= \frac{g^2m_t^3}{256\pi^2m_W^2}\int\frac{p_b\mathrm{d}p_b\mathrm{d}\Omega}{m_t-p_b}\delta\left(p_b-\frac{m_t}{2}\right)
\end{aligned}$$

$$\boxed{\Gamma_t \simeq \frac{g_W^2 m_t^3}{64\pi m_W^2}}$$

Numerically, using $m_t = 172.69$ GeV, $m_W = 80.377$ GeV, and $g_W^2 = 4\sqrt{2}\,G_F m_W^2$:

$$\Gamma_t \simeq 1.49 \text{ GeV}$$

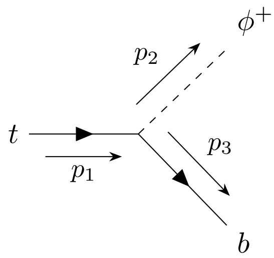
which compares well with the PDG value $\Gamma_{t,\text{PDG}} = 1.42^{+0.19}_{-0.15}$ GeV.

This decay is enhanced in the limit $m_t/m_W \gg 1$. Using the equivalence theorem, which states that the amplitude for emission of a longitudinal W equals the amplitude for emission of the corresponding Goldstone boson ϕ^+ at high energies, we expect Γ_t to be governed by the top Yukawa coupling λ_t. Replacing the masses $m_t = \lambda_t v/\sqrt{2}$ and $m_W = g_W v/2$ we expect

$$\Gamma_t \simeq \frac{g_W^2 m_t^3}{64\pi m_W^2} = \frac{\lambda_t^2 m_t}{32\pi}$$

We now verify this by computing the decay of t into a Goldstone boson and b. The coupling of the charged Goldstone ϕ^+ to the top-bottom pair arises from the Yukawa interaction, giving a vertex $\propto \lambda_t P_R$ (neglecting m_b).

$$t \longrightarrow \phi^+ b$$



$$\mathcal{M} = \frac{\lambda_t}{2}\,\bar{u}_3(1+\gamma^5)u_1 \qquad \Longrightarrow \qquad \langle|\mathcal{M}|^2\rangle = \lambda_t^2(p_1\cdot p_3) \simeq \frac{\lambda_t^2 m_t^2}{2}\left(1 - \frac{m_W^2}{m_t^2}\right) \simeq \frac{\lambda_t^2 m_t^2}{2}$$

$$\boxed{\Gamma(t\to\phi^+ b) \simeq \frac{\lambda_t^2 m_t}{32\pi}}$$

This confirms the equivalence theorem: in the limit $m_t/m_W \gg 1$, the W^+ in the final state is dominantly longitudinal, and the decay rate is controlled by the Yukawa coupling λ_t rather than the gauge coupling g_W.

<u>Electron stability in the limit $g_W \to 0$:</u>

Now we consider an imaginary world with $g_W \to 0$ but the same Higgs expectation value $v = 246$ GeV. Since $m_W = g_W v/2$, the W boson becomes massless in this limit ($m_W \to 0$). Therefore the decay $e^- \to W^-\nu_e$, which is normally kinematically forbidden because $m_e \ll m_W$, becomes kinematically allowed. By analogy with $t \to W^+b$, the rate would be

$$\Gamma_e = \frac{g_W^2 m_e^3}{64\pi m_W^2} = \frac{m_e^3}{16\pi v^2}$$

where in the last step we substituted $m_W = g_W v/2$. Crucially, the factors of g_W cancel, and Γ_e is **finite and non-zero** in the limit $g_W \to 0$:

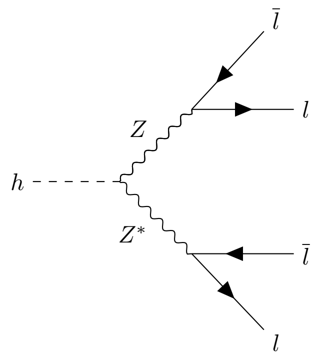
$$\Gamma_e = \frac{m_e^3}{16\pi v^2} \simeq 2.3 \times 10^{-18} \text{ GeV}$$

The electron is therefore **unstable** in this limit. This can again be understood via the equivalence theorem: the electron decays into a Goldstone boson ϕ^- and ν_e via its Yukawa coupling $\lambda_e = \sqrt{2}\, m_e/v$, which persists even as $g_W \to 0$.

1.5 Decay Width of 125 GeV SM Higgs

Calculate the decay width of the 125 GeV SM Higgs boson to four "stable" lepton pairs (electrons and muons), mediated by ZZ^* intermediate states (one on-shell, and one off-shell Z boson), and to $W^\pm$ + 2 light particles final states mediated by WW^*. For the three-body final states, use the PDG review on kinematics, and obtain first the decay rate as a function of the invariant masses, m_{12} and m_{23}. One integration can be performed analytically, while the second one you may perform numerically. Examine your answers by comparing with observations/known SM predictions.

$$h \longrightarrow ZZ^* \longrightarrow 4l$$



Let p_1 and p_2 be the momenta of the fermions from the off-shell Z^*, and p_3 be the momentum of the on-shell Z. Using the standard 3-body phase space formula from the PDG kinematics review, the differential decay rate is given by:

$$\mathrm{d}\Gamma(h \to ZZ^* \to Zl\bar{l}) = \frac{1}{(2\pi)^3}\frac{1}{32m_h^3}\overline{|\mathcal{M}|^2}\,\mathrm{d}m_{12}^2\,\mathrm{d}m_{23}^2$$

where $\overline{|\mathcal{M}|^2}$ contains the off-shell Z^* propagator $1/(m_{12}^2 - m_Z^2)^2$. The integration limits for m_{23}^2 at a fixed m_{12}^2 are determined by the reaction kinematics:

$$(m_{23}^2)_{\max,\min} = (E_2^* + E_3^*)^2 - \left(\sqrt{E_2^{*2} - m_2^2} \mp \sqrt{E_3^{*2} - m_3^2}\right)^2$$

where E_2^* and E_3^* are the energies of particles 2 and 3 in the m_{12} rest frame. Integrating over m_{23}^2 analytically leaves a 1D integral over m_{12}^2, which is bounded between 0 and $(m_h - m_Z)^2$. Evaluating this numerically for the electrons and muons channels yields (see `task-1-5.m`):

$$\Gamma(h \to 4l) = \Gamma(h \to Z^*\bar{l}l) \times \frac{\Gamma(Z \to \bar{l}l)}{\Gamma_Z} \simeq 3.78 \times 10^{-7}\ \text{GeV}$$

Given the total Higgs width $\Gamma_h \simeq 4.07$ MeV, the tree-level branching ratio is:

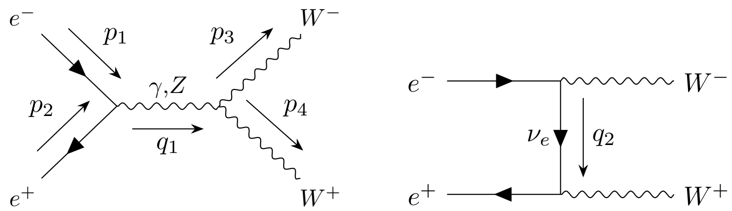
$$\mathrm{Br}(h \to 4l) = \frac{\Gamma(h \to 4l)}{\Gamma_h} \simeq 0.928 \times 10^{-4}$$

The CERN Yellow Report (2016) SM prediction for $\mathrm{Br}(h \to 4l)$ is $\simeq 1.24 \times 10^{-4}$. Our tree-level calculation captures roughly $\sim 75\,\%$ of the total rate. This discrepancy is expected because our calculation assumes the Narrow Width Approximation (NWA) for the "on-shell" Z. For a 125 GeV Higgs, $m_h < 2m_Z$, meaning the decay is deeply in the threshold region where both Z bosons should ideally be treated with full Breit-Wigner distributions.

1.6 Cross Section of $e^+e^- \to W^+W^-$

Calculate the cross section for $e^+e^- \to W^+W^-$ pair production at LEP II by considering the strength of the $W-W-Z$ coupling to be a floating parameter. It is of course fixed by the requirement of the $SU(2)$ gauge invariance. Does the comparison of this cross section with experiment (see, *e.g.* hep-ex/0506048) "recovers" the correct value for the triple gauge vertex? The full SM analytic expression for the cross section is given in (51.26) of PDG. If you are stuck in your calculations, use expression (51.26) to integrate over t and compare with experiment.

$$e^+e^- \longrightarrow W^+W^-$$



Matrix Elements:

$$\begin{aligned}
\mathcal{M}_\gamma &= -\frac{e^2}{q_1^2}(\bar{v}_2\gamma_\lambda u_1)[g^{\mu\nu}(p_3-p_4)^\lambda + g^{\lambda\nu}(-q_1-p_3)^\mu + g^{\lambda\mu}(q_1+p_4)^\nu]\varepsilon^*_{\mu 4}\varepsilon^*_{\nu 3} \\
\mathcal{M}_Z &= \frac{g_W g_{wwz}}{2\cos\theta_w(q_1^2-m_Z^2)}[\bar{v}_2\gamma_\lambda(g_V - g_A\gamma_5)u_1][g^{\mu\nu}(p_3-p_4)^\lambda + g^{\lambda\nu}(-q_1-p_3)^\mu + g^{\lambda\mu}(q_1+p_4)^\nu]\varepsilon^*_{\mu 4}\varepsilon^*_{\nu 3} \\
\mathcal{M}_\nu &= -\frac{g_W^2}{4}\bar{v}_2\gamma^\mu\frac{\not{q}_2}{q_2^2}\gamma^\nu(1-\gamma_5)u_1\varepsilon^*_{\mu 4}\varepsilon^*_{\nu 3}
\end{aligned}$$

$$|\mathcal{M}|^2 = |\mathcal{M}_\gamma|^2 + |\mathcal{M}_Z|^2 + |\mathcal{M}_\nu|^2 + 2\,\mathrm{Re}(\mathcal{M}^*_\gamma\mathcal{M}_Z + \mathcal{M}^*_Z\mathcal{M}_\nu + \mathcal{M}^*_\gamma\mathcal{M}_\nu)$$

where the triple gauge boson vertex $V^{\mu\nu\lambda}(p_3,p_4,q_1) = g^{\mu\nu}(p_3-p_4)^\lambda + g^{\lambda\nu}(-q_1-p_3)^\mu + g^{\lambda\mu}(q_1+p_4)^\nu$.

CM Frame (γ, Z):

$$p_1 = (E,\vec{p}_1)\,; \qquad p_2 = (E,-\vec{p}_1)\,; \qquad p_3 = (E,\vec{p}_3)\,; \qquad p_4 = (E,-\vec{p}_3)$$
$$q_1^2 = 4E^2 = E_{cm}^2\,; \quad p_1\cdot p_3 = p_2\cdot p_4 = E^2 - |\vec{p}_1||\vec{p}_3|\cos\theta\,; \quad p_1\cdot p_4 = p_2\cdot p_3 = E^2 + |\vec{p}_1||\vec{p}_3|\cos\theta$$

Cross-section:

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{cm} = \frac{p_f}{32\pi^2 E_{cm}^3}|\mathcal{M}|^2$$

(see `task-1-6.m`)